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Enhancement of dilution and transverse reactive mixing in porous media: Experiments and model-based interpretation

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ABSTRACT

Transport and natural attenuation of contaminant plumes in groundwater are often controlled by transverse dispersion. The extent of mixing between dissolved reaction partners at the fringe of a plume determines its length and depends strongly on the groundwater flow field. Transient flow conditions as well as the focusing of the flow in high-permeability zones may enhance transverse mixing of dissolved species and, therefore, create favorable conditions for the natural attenuation of contaminant plumes. The aim of the present study is to experimentally test the influence of these processes on solute mixing and to directly compare the results with those under analogous homogeneous and steady-state conditions. We have performed conservative and reactive tracer experiments in a quasi two-dimensional tank filled with glass beads of different sizes. The experiments have been carried out in both homogeneous and heterogeneous porous media under steady-state and transient (i.e. oscillating) flow fields. We used fluorescein as conservative tracer; whereas an alkaline solution (NaOH) was injected into ambient acidic water (HCl) in the reactive experiments. A pH indicator was added to the reacting solutions in order to visualize the emerging plume. We simulated the laboratory experiments with a numerical model and compared the outcomes of the model with the measured concentrations at the outlet of the tank and with the observed tracer plumes. Spatial moments, a newly defined flux-related dilution index, the product mass fluxes and the reaction enhancement factors were calculated to quantify the differences in mixing and reaction extent under various experimental conditions. The results show that flow focusing in heterogeneous porous media significantly enhances transverse mixing and mixing-controlled reactions, whereas temporally changing flow fields appear to be of minor importance.

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1. Introduction

Mass transfer, mixing and hence reaction rates during transport of solutes in groundwater depend on dispersion processes. When contaminant plumes evolving from continuous sources approach steady-state conditions, vertical transverse dispersion controls the mixing of reaction partners (i.e. electron donors and acceptors for microbial degradation of organic contaminants) thus determining the effective reaction rates and the plume length (e.g. Cirpka et al., 1999; Mayer et al., 2001; Ham et al., 2004; Liedl et al., 2005; Maier and Grathwohl, 2006). Pore-scale mixing of the reaction partners is a necessary condition for the reaction between two or more dissolved compounds (Kitanidis, 1994; Cirpka, 2002). Experimental and modeling investigations at different scales showed that vertical transverse dispersion coefficients are typically very small (e.g. Sudicky, 1986; Klenk and Grathwohl, 2002, Cirpka et al., 2006; Olsson and Grathwohl, 2007). As a consequence, mixing-controlled reactions of dissolved compounds can only take place in narrow stripes at the fringes of contaminant plumes. Evidence that important reactive processes are restricted to the fringe of contaminant plumes were given both at the laboratory (Huang et al., 2003; Willingham, 2006; Cirpka et al., 2006; Rees et al.,

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2007; Rolle et al., 2008a; Willingham et al., 2008; Bauer et al., 2008, 2009a, 2009b) and at the field scale (Davis et al., 1999; Lerner et al., 2000; Tuxen et al., 2006; Rolle et al., 2008b; Anneser et al., 2008, Prommer et al., in press). In detailed field investigations of an aquifer contaminated by phenolic compounds, Lerner et al. (2000) showed the occurrence of fringe degradation reactions between the plume contaminants and the dissolved electron acceptors in the background groundwater. Similar findings were obtained at a landfill site (Tuxen et al., 2006) and, recently, at a site contaminated by petroleum hydrocarbons (Anneser et al., 2008). The latter study showed, by means of high resolution multi-level sampling devices, the presence of narrow (mm–cm) reactive zones and the occurrence of steep concentration gradients of reactive biogeochemical species at the plume fringe.

All these studies show that mixing processes are of utmost importance in determining the fate and transport of contaminants in groundwater and they often represent the limiting factor to the overall degradation. For these reasons considerable attention should be dedicated to the investigation of processes possibly enhancing the mixing of reactants in porous media, in particular in the transverse direction.

Transient flow conditions have been indicated as a possible source of mixing enhancement. For transport of non-reactive chemicals in homogeneous porous media, it was shown (Kinzelbach and Ackerer, 1986; Goode and Konikow 1990) that fluctuations in the groundwater flow direction due to transient conditions can lead to an increase of transverse dispersion. In modeling studies of organic contaminant plumes, Schirmer et al. (2001) and Prommer et al. (2002) investigated the influence of transient flow conditions on contaminant transport and biodegradation in homogeneous porous aquifers. Both studies found just slightly increased transverse dispersion and therefore degradation enhancement. The latter authors suggested that the presence of transient flow combined with sorption processes could lead to a more significant mixing enhancement. Similar conclusions were obtained by Cirpka (2005) for solute transport and kinetic sorption in a spatially homogeneous flow field undergoing sinusoidal fluctuations in time.

Many studies based on both deterministic and stochastic approaches, investigated the importance of sediment heterogeneities in determining the flow field in saturated porous media and, therefore, the transport of dissolved compounds. Linear stochastic theory predicts only small enhancement of transverse effective dispersion by heterogeneity (Dentz et al., 2000; Fiori and Dagan, 2000). This theory results in an enhancement of transient mixing in heterogeneous media of approximately one order of magnitude under transient flow conditions (Cirpka and Attinger, 2003; Dentz and Carrera, 2003). However, the appropriateness of linear stochastic theory for transverse dispersion may be questioned (Attinger et al., 2004), since higher-order effects such as the enhancement of transverse mixing by flow focusing (Werth et al., 2006) may be dominant. Experimental studies of transverse mixing in heterogeneous media are scarce, and the interpretation of individual transverse concentration profiles may be impeded by the deformation of the flow field (Rahman et al., 2005).

In this study, laboratory bench-scale experiments were performed, in quasi two-dimensional flow-through systems, to deepen the understanding of vertical transverse dispersion in porous media and to quantify the influence of transient flow fields and heterogeneities on mixing and reactions. We carried out transport experiments in saturated unconfined porous media under steady-state and transient flow fields using both conservative and reactive tracers. The same experimental setup was used for both uniform and heterogeneous porous media, thus allowing a direct comparison between the different experiments. In the latter experiments, the heterogeneities consisted of two well-defined high-permeability zones (coarse material) embedded into a lower permeability matrix.

Numerical simulations of the conservative and reactive tracer experiments may help interpreting the laboratory results. We compared model results with the measured concentrations at the outlet of the tank and with the observed conservative and reactive color plumes. A quantitative evaluation of the experimental and modeling results was performed through the calculation of spatial moments, dilution indexes, mass fluxes of reaction product and reaction enhancement factors.

2. Material and methods

2.1. Experimental setup

The experiments were carried out in a quasi 2-D acrylic-glass flow-through tank with inner dimensions of 77.9 cm length, 15.0 cm height and 1.1 cm width (Fig. 1). The tank was equipped with 13 equally spaced ports (1.1 cm distance) as inlets and outlets on both sides. The ports were made of pierced rubber septa, and the injection needles were directly connected to tygon pump-tubing (ID 0.57 mm, Ismatec, Glattbrugg, Switzerland). For the experiments performed under steady-state flow conditions, two 12-channel high precision peristaltic pumps (IPC-N, Ismatec, Glattbrugg, Switzerland), one for the inlet ports and the other for the outlet ports, were operated at identical pumping rates.

The tank was filled with glass beads up to a height of approximately 14 cm. The top of the domain was kept unsaturated to avoid an upper shortcut of the injected tracer solutions. In the experiments with a homogeneous filling, we used glass beads of uniform grain size of 0.25–0.3 mm (Sartorius AG, Goettingen, Germany).

Transient flow conditions in the homogeneous porous medium setup were established by using three peristaltic pumps. While the inlet ports 2–8 were connected to pump 1 and operated with a flow rate of $q_{port} = 7.083 \times 10^{-4}$ mL/s for each port, inlet port 1 was connected to pump 2 and set to the double flow rate ($2 \times q_{port}$) and inlet port 10 was connected to pump 3 set to no flow rate ($q_{port} = 0$). At the outlet, the water was withdrawn from the uppermost 5 ports, which were connected to pump 2 with the double flow rate ($2 \times q_{port}$), while the lower 5 ports were connected to pump 3. At a subsequent stress period the flow rates were changed symmetrically: the flow rate of inlet port 1 and the outlet ports 6–10 were set to 0, while the ports connected to pump 3 (inlet port 10 and the outlet ports 1–5) were operated at a double flow rate ($2 \times q_{port}$).

To create a heterogeneous porous medium, two inclusions of higher permeability made of glass beads of a grain size of 1–1.5 mm (Fisher Scientific GmbH, Schwerte, Germany) were embedded into the finer glass beads (0.25–0.3 mm). The two high-permeability zones were placed directly in line with the

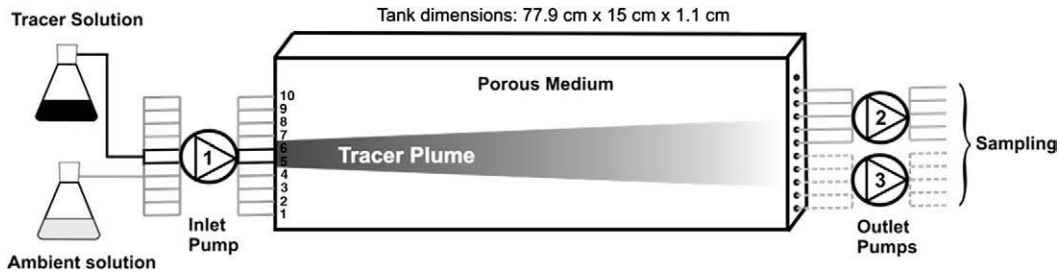

Fig. 1. Experimental setup.

central plume's injection port and their thickness was equal to the port spacing (1.1 cm). The length of the inclusions was 20 cm, the distance of the first zone from the injection port was 15 cm and the distance between the two inclusions was 8 cm.

In the experiments with conservative color tracer, a sodium fluorescein solution with a concentration of 30 mg/L was injected through the central inlet port (port 6), while clean water was injected through the other ports to establish parallel flow. After the displacement of two pore volumes, when the plume reached steady-state conditions, samples were taken at each outlet port and the fluorescence signal was measured using a fluorescence spectrometer (Perkin Elmer LS-3B). Conservative experiments were repeated under transient flow conditions in the homogeneous porous medium.

In the reactive experiments, a solution of 0.004 mol/L NaOH (pH = 11.49) was injected through two middle inlet ports into an ambient solution of 0.01 mol/L HCl (pH = 2.03), which was injected through the surrounding ports. Both solutions were prepared from deionized water and contained the pH indicator bromophenol blue at a concentration of 5.92×10^{-6} mol/L. The indicator changes its color from yellow to blue in the pH interval of 3–4.6. In the heterogeneous setup, the reactive experiments were repeated with increasing concentrations of the alkaline solution (0.008, 0.022 and 0.030 mol/L). Pictures of the alkaline plume were taken at different times in both the experiments carried out under steady-state and transient conditions.

2.2. Hydraulic properties

The most important parameters determining the flow field in the tank systems (i.e. hydraulic conductivity, porosity) and the transport of conservative and reactive solutes (i.e. longitudinal and transverse dispersivities) were determined experimentally and are summarized in Table 1.

Permeameter tests were conducted to determine the hydraulic conductivities of the fillings with different grain sizes used as porous media in the two-dimensional flow-through systems, resulting in average values of 6.14×10^{-4} m/s and 1.27×10^{-2} m/s for the fine (0.25–0.3 mm) and coarse (1–1.5 mm) glass beads, respectively. Thus, the ratio of the hydraulic conductivities between the two different porous media in the heterogeneous system was 20.7. This parameter determines the amount of water flow that is focused in the two high-permeability inclusions of the heterogeneous setup and therefore plays a very important role for the mixing enhancement in the heterogeneous system compared to the equivalent homogeneous setup (Werth et al., 2006).

Transverse dispersivities for the two different grain sizes were determined in separate homogeneous tank experiments with fluorescein by measuring the tracer distribution at the outlet ports. In the tank filled with fine glass beads an average linear velocity of 1.1 m/day was established, whereas in the tank with the coarser grain size a higher value of linear velocity (10 m/day) was set in order to closely match the expected conditions in the high-permeability inclusions of the heterogeneous setup. In these homogeneous systems the simplifying assumption of uniform parallel flow could be considered valid. The simplified two-dimensional analytical solutions of the transport equation for a line source under steady-state (1) and transient conditions (2) (Domenico and Palciauskas, 1982; Domenico and Robbins, 1985) could be applied since the constraints pointed out by Srinivasan et al. (2007) are satisfied:

$$\frac{C(x,y)}{C_0} = \frac{1}{2}\left\{erf\left[\frac{(y+Y/2)}{2(xD_T/v)^{1/2}}\right] - erf\left[\frac{(y-Y/2)}{2(xD_T/v)^{1/2}}\right]\right\} \quad (1)$$

$$\frac{C(x,y,t)}{C_0} = \frac{1}{4}erfc\left[\frac{(x-vt)}{2(D_L t)^{1/2}}\right] \times\left\{erf\left[\frac{(y+Y/2)}{2(xD_T/v)^{1/2}}\right] - erf\left[\frac{(y-Y/2)}{2(xD_T/v)^{1/2}}\right]\right\} \quad (2)$$

where D_T and D_L [L^2T^{-1}] are the transverse and longitudinal hydrodynamic dispersion coefficients, C [ML^{-3}] is the tracer concentration, C_0 [ML^{-3}] is the tracer concentration at the source, Y [L] is the source width, y [L] is the spatial coordinate in lateral direction, x [L] is the spatial coordinate in longitudinal direction, t [T] is the time and v [LT^{-1}] is the pore water velocity. In order to determine the transverse dispersion coefficient D_T, the analytical solution of Eq. (1) was fitted to the fluorescein concentrations measured at the outlet ports, minimizing the sum of the squared residuals. Successively, the transverse dispersivity, α_T [L], was calculated according to the standard linear model for saturated porous media (Scheidegger, 1961):

$$\alpha_T = \frac{D_T - D_P}{v} \quad (3)$$

in which the effective pore diffusion coefficient D_P [L^2T^{-1}] can be approximated as $D_P = D_{aq}n$ (Grathwohl, 1998; Boving and Grathwohl, 2001), where D_{aq} is the aqueous diffusion coefficient [L^2T^{-1}] and n [–] is the porosity. A value of $\alpha_T = 1.4\times10^{-5}$ m was determined for the glass beads with 0.25–0.3 mm diameter at a seepage velocity $v = 1.1$ m/day, and $\alpha_T = 5.1\times10^{-5}$ m for

Table 1
Summary of flow and transport parameters.

Parameter	Value
Tank inner dimension (m)	$0.779 \times 0.15 \times 0.011$
Average fine grain diameter (mm)	0.275
Average coarse grain diameter (mm)	1.25
Number of inlet/outlet ports used (–)	10
Total flow rate (mL/s)	7.083×10^{-3}
Average linear pore water velocity (m/s)	1.27×10^{-5}
Porosity fine grain (–)	0.42
Porosity coarse grain (–)	0.40
Hydraulic conductivity fine material (m/s)	6.14×10^{-4}
Hydraulic conductivity coarse material (m/s)	1.27×10^{-2}
Hydraulic conductivity ratio (–)	20.7
Longitudinal dispersivity, α_L fine material (mm)	0.183
Longitudinal dispersivity, α_L coarse material (mm)	0.767
Transverse dispersivity, α_T fine material (mm)	0.014
Transverse dispersivity, α_T coarse material (mm)	0.051
Effective pore diffusion coefficient for fluorescein (m^2/s)	2.72×10^{-10}
Effective pore diffusion coefficient for NaOH and HCl solutions (m^2/s)	1.41×10^{-9}

the coarse glass beads (1–1.5 mm) at $v = 10$ m/day. To determine the longitudinal dispersion coefficient D_L the analytical solution of Eq. (2) was fitted to the measured breakthrough concentrations of fluorescein at the central outlet port.

Successively, longitudinal dispersivities were calculated as:

$$\alpha_L = \frac{D_L - D_P}{v}. \quad (4)$$

A value of $\alpha_L = 1.83 \times 10^{-4}$ m was obtained for the fine grain size, whereas $\alpha_L = 7.67 \times 10^{-4}$ m was determined for the coarse glass beads.

Porosity values of 0.42 and 0.40 were calculated from the specific discharge and the mean arrival time of the tracer in the homogeneous tank system for the fine and coarse grain sizes, respectively.

2.3. Numerical modeling

In order to analyze the experimental results, we simulated flow and transport in the different flow-through systems with a numerical model. The model domain was discretized into 35 530 cells with varying spacing ($\Delta x_{min} = 1 \times 10^{-3}$ m, $\Delta x_{max} = 5 \times 10^{-3}$ m) along the flow direction and a constant spacing ($\Delta z = 7.5 \times 10^{-4}$ m) in the vertical direction.

The flow field was simulated using MODFLOW (McDonald and Harbaugh, 1988) and particle tracking simulations were performed using MODPATH (Pollock, 1994). The flow-through systems were simulated as unconfined aquifers using the hydraulic parameters reported in Table 1. Neumann boundary conditions were applied and active cells, with a distance equivalent to the ports spacing in the experimental setup, were used to simulate the inlet and outlet ports.

Non-reactive tracer experiments were simulated using MT3DMS (Zheng and Wang, 1999) with a third order total variation diminishing (TVD) scheme to solve the advection problem and an implicit finite difference scheme for the remaining terms of the advection–dispersion equation.

The reactive transport simulations of the acid/base systems were carried out with the multi-component reactive transport code PHT3D (Prommer et al., 2003), which combines MT3DMS (Zheng and Wang, 1999) with the geochemical model PHREEQC-2 (Parkhurst and Appelo, 1999).

3. Results and discussion

3.1. Conservative transport experiments

3.1.1. Steady-state plumes

In the homogeneous system the steady-state conservative tracer plume is characterized by a thin lateral extension which broadens with increasing distance from the inlet

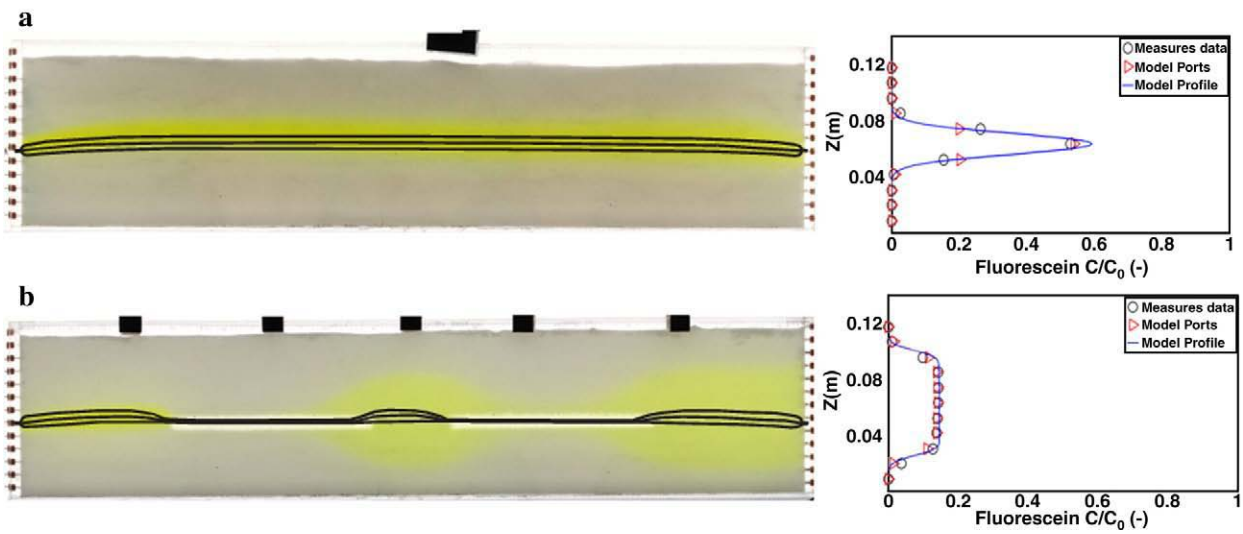

Fig. 2. Steady-state fluorescein plumes in the homogeneous (a) and heterogeneous (b) tank and computed flow lines. On the right, measured triplicate fluorescein concentrations, normalized to the inlet concentration $C_0 = 30$ mg/L (circles), simulated results at the outlet ports (triangles) and a vertical profile inside the tank 2 cm from the outlet (solid line).

(Fig. 2a). The picture also shows low color intensity at the plume fringe and close to the outlet of the tank, corresponding to a decrease of tracer concentration as a result of dispersive processes. The fluorescein concentrations measured at the outlet ports show a characteristic Gaussian-type shape. The plume shape is quite different in the heterogeneous system. Here the permeability contrast between the two different porous media induces focusing of flow into the high-permeability inclusions. This behavior is shown in Fig. 2b by the computed flowlines which converge within the coarser inclusions. In the homogeneous tank the observed plume is just moderately thicker than the region occupied by the computed streamlines originating from the central inlet reflecting the plume boundary in purely advective transport. This difference can be attributed to the contribution of transverse dispersion to the plume spreading in the homogeneous porous medium at a velocity of 1.1 m/day. On the contrary, in the heterogeneous setup, the small distance among the streamlines in the high-permeability inclusions favors the diffusive exchange of solute mass and enhances transverse mixing. Therefore, the fluorescein plume close to the outlet of the heterogeneous setup appears to be spread over almost the complete aquifer thickness and exhibits lower concentrations, as shown in the profiles reported in Fig. 2b. For both the homogeneous and heterogeneous setups a good agreement between observed and simulated results was obtained. Triplicate measurements of fluorescein at the outlet ports of the tanks are compared with the results of the numerical transport model at the virtual outlet ports and with simulated vertical profiles taken inside the porous medium at 2 cm from the right boundary of the homogeneous and heterogeneous flow-through systems.

3.1.2. Moment analysis of transient plumes

Transient flow conditions in the homogeneous porous medium setup were established by using an additional peristaltic pump and resulted in oscillating fluorescein plumes. Snapshots at different times were taken for two experiments corresponding to a different frequency of flow oscillations (Fig. 3). Fig. 3a shows the results at different times ($t_1 = 2.82$, $t_2 = 2.94$, and $t_3 = 3.00$ days) for an oscillating period of half the travel time of the solute in the tank and Fig. 3b shows the plumes corresponding to an oscillation period of a quarter of the travel time for $t_1 = 0.71$, $t_2 = 0.78$ and $t_3 = 0.88$ days. In the same figure, the corresponding outcomes of conservative transport simulations are reported.

We computed the spatial moments of the corresponding simulated concentrations. In a three-dimensional saturated

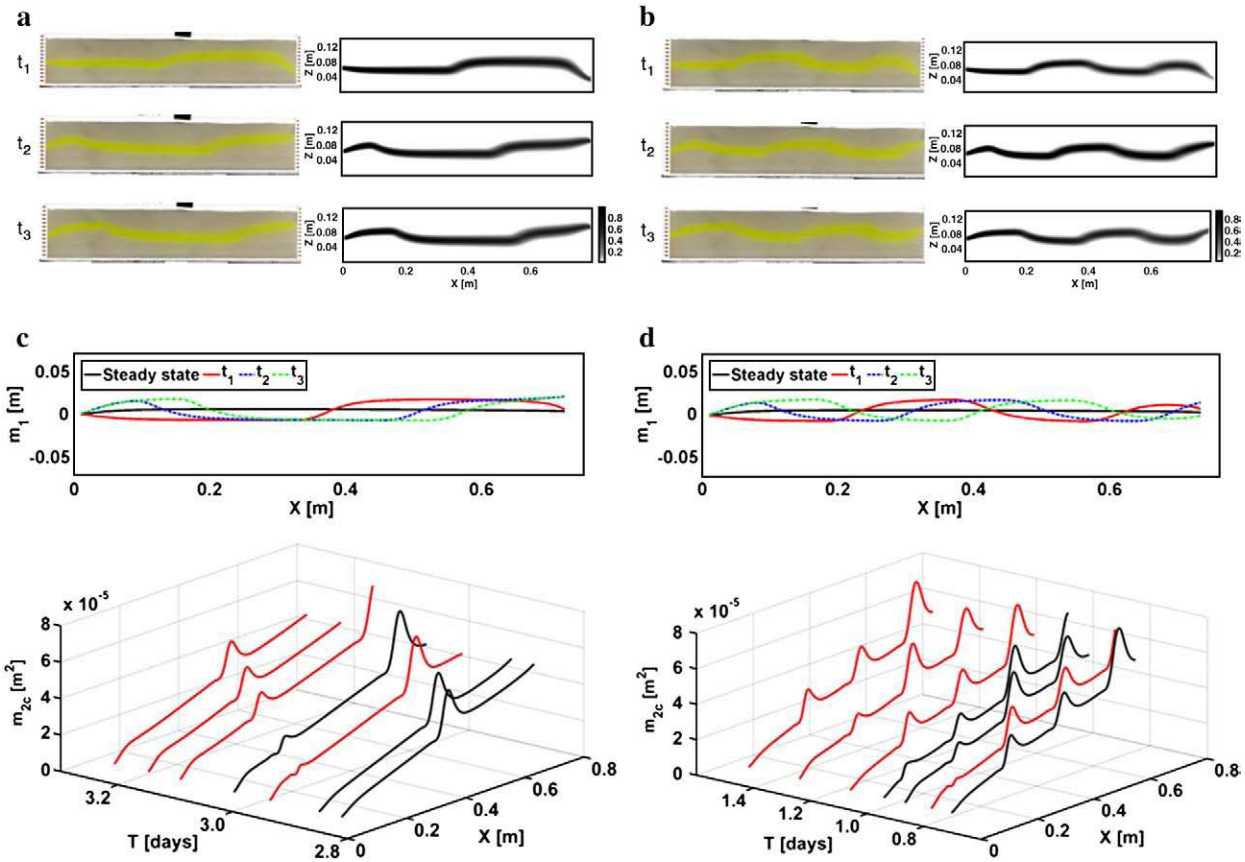

Fig. 3. Observed and simulated fluorescein plumes corresponding to changes of boundary conditions of half- and a quarter of the solute travel time in the tank (a and b). Computed first and second central spatial moments in the homogeneous system (c and d). In the plots of the second central spatial moments the black solid lines correspond to the time of the pictures.

porous medium, the raw spatial moments of the concentration distribution *C(x,y,z,t)* are defined as (Aris, 1956):

$$M_{ijk}(t) = \int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} nC(x,y,z,t)x^i y^j z^k dx dy dz \quad (5)$$

where n represents the porosity and x, y and z the spatial coordinates. We calculated the zeroth, first and second spatial moments of the fluorescein distribution for the steady-state experiment and for the transient setups at the times for which the pictures and the outcomes of the numerical modeling were available. In our application, the concentration distribution is uniform over the width of the domain (W_y). Also, the size of the cross-section (L_z) is limited so that the integration in the lateral coordinates is bounded. The first moment normalized by the zeroth one represents the location of the center of gravity of the solute plume. For the continuous-injection experiments, the normalized first spatial moment in the vertical direction was calculated as:

$$m_1(x) = \frac{\int_0^{W_y}\int_0^{L_z} z\cdot c(x,y,z)dydz}{\int_0^{W_y}\int_0^{L_z} c(x,y,z)dydz} \quad (6)$$

which is computed for each column of the computational domain.

The results corresponding to the different times shown in Fig. 3a and b are reported in Fig. 3c and d. The normalized first moments of the transient plumes change their position in comparison to the steady-state fluorescein plume (black solid line) with the same phase as applied in the flow boundary conditions.

The normalized second central spatial moment is a measure of the spreading of the concentration distribution about its center of gravity and its value in the vertical transverse direction was calculated as:

$$m_{2C}(x) = \frac{\int_0^{W_y}\int_0^{L_z} (z-m_1(x))^2\cdot c(x,y,z)dydz}{\int_0^{W_y}\int_0^{L_z} c(x,y,z)dydz}. \quad (7)$$

Fig. 3c and d shows the normalized second central moments for each vertical section along the length of the tank at different times. Under transient conditions the second central spatial moments increase where the plume bends. In fact, in these zones, the longitudinal dispersion is acting in a direction that is no more horizontal but presents a significant component also in the vertical transverse direction (Kinzelbach and Ackerer 1986; Goode and Konikow, 1990). Several authors (e.g. Kitanidis 1994, Cirpka 2002) pointed out the difference between plume spreading and mixing, with the former concept related to the change of the plume shape whereas the latter referred to the change of volume occupied by the solute. However, in the homogeneous flow-through setup, both plume spreading and dilution are described by local dispersion coefficients, and the computed second central spatial moments increase with time (Fig. 3c and d). These findings suggest that the second central spatial moments, representing the fluorescein plume spreading, can be assumed as a good proxy for quantifying plume mixing in the homogeneous system. This conclusion clearly does not apply for heterogeneous porous media, where plume spreading does not represent a measure of mixing (or dilution), as can be observed in Fig. 4. The plot shows the computed second central transverse moments in the homogeneous and heterogeneous systems under steady-state flow conditions. When the plume is focused inside the high-permeability inclusions, its second central transverse moment decreases while its actual dilution cannot decrease because dilution is an irreversible process.

3.1.3. Quantification of mixing enhancement by the dilution index

Since second central spatial moments are inadequate to quantify plume mixing and dilution in heterogeneous porous media, another measure of these processes is required in order to perform a quantitative analysis of the experimental results. We refer here to the concept of dilution index introduced by Kitanidis (1994). In its original form, the dilution index is defined as:

$$E(t) = exp\left[-\int_V p(\vec{x},t)ln(p(\vec{x},t))\mathrm{d}V\right] \quad (8)$$

where $p(\vec{x},t)$ is the distribution function of the solute (concentration normalized by the total mass of the solute in the domain).

The illustrative explanation of the original dilution index is the effective volume occupied by a solute cloud. This is a useful property for the analysis of mixing when the compound is introduced as an individual slug, and when the concentration distribution in the entire domain is observable at any given time of interest. However, the original dilution index gives no indication how the dilution of the solute increases with the travel distance when the compound is continuously injected into the flow domain, as occurred in our experimental setups. Therefore, we define a modified dilution index that can be obtained in a cross-

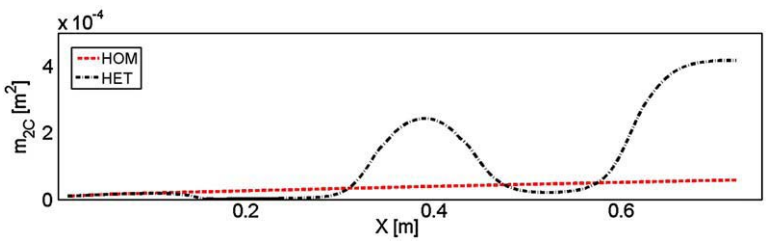

Fig. 4. Normalized second central spatial moments in the transverse direction computed for the homogeneous and heterogeneous setups.

section perpendicular to the mean direction of flow and expresses an effective discharge over which the solute mass flux is distributed.

In the following, the reference framework is that of solute flux. While in the discussion of the original volume-related dilution index, dilution was defined as "the act of distributing solute mass over a larger volume" (Kitanidis, 1994), we address dilution as the act of distributing a given solute mass flux over a larger water flux. The concentrations considered are local flux concentrations, that is, the solute mass flux in the direction normal to the control plane divided by the specific discharge in that direction. As is well known, the difference between flux and resident concentration lies in the normal gradient component of the resident concentration times the bulk dispersivity (Kreft and Zuber, 1978). At the local scale, this difference is negligible in advection dominated transport.

Computing the dilution index involves integration. In the volume-related dilution index, this was an integral over the volume of the domain. In our flux-related framework, we compute the integral over the water flux which can be expressed as integral of the quantity to be integrated times the normal component of the local specific discharge over the cross-sectional area.

We start the analysis by considering the total solute mass flux:

$$F(x) = \int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} c(x,y,z)\cdot q_x(x,y,z)dydz \tag{9}$$

where $c(x,y,z)$ is the flux-related concentration and $q_x(x,y,z)$ is the normal component of the specific discharge with respect to a cross-sectional area in the plane y–z.

A flux-related density function of the solute mass can be defined as the ratio between the concentration and the mass flux:

$$p_Q(x,y,z) = \frac{c(x,y,z)}{F(x)} \tag{10}$$

which has units of an inverse discharge, and expresses the probability that a solute particle, in the cross-section at distance x, is within a certain fraction of the water flux. It can easily be shown that, p_Q is a true flux-related density function since its integral over the whole water flux crossing the control plane is unity:

$$\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} p_Q(x,y,z)\cdot q_x(x,y,z)dydz = 1. \tag{11}$$

Then, the flux-related dilution index becomes:

$$E_Q(x) = exp\left[-\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} p_Q(x,y,z)\cdot ln(p_Q(x,y,z))\cdot q_x(x,y,z)dydz\right] \tag{12}$$

which has units of a discharge.

$E_Q(x)$ quantifies an effective volumetric flux transporting the solute flux at the longitudinal position x. In a conservative system at steady-state $E_Q(x)$ can only increase with travel distance because transverse dispersion distributes the solute flux over a larger water flux. In contrast, the cross-sectional area over which the solute flux is transported may decrease in regions of converging flow.

In our application, the integration in the lateral coordinates is bounded, and the flux-related dilution index can be written as:

$$E_Q(x) = exp\left[-\int_0^{W_y}\int_0^{L_z} p_Q(x,y,z)\cdot ln(p_Q(x,y,z))\cdot q_x(x,y,z)dydz\right] \tag{13}$$

In a system with finite cross-section, the upper limit of the flux-related dilution index $E_Q(x)$ is the total discharge passing through the system:

$$E_Q^{max}(x) = Q = \int_0^{W_y}\int_0^{L_z} q_x(x,y,z)dydz \tag{14}$$

in which $E_Q(x) = E_Q^{\max}(x)$ requires a uniform concentration distribution over the cross-section.

In analogy to the volume-related dilution index of Kitanidis (1994), we may define a flux-related reactor ratio $M_Q(x)$, by normalizing $E_Q(x)$ with $E_Q^{\max}(x)$:

$$M_Q(x) = \frac{E_Q(x)}{E_Q^{max}(x)} \tag{15}$$

which cannot exceed the value of unity.

Fig. 5 shows the computed flux-related dilution indices $E_Q(x)$ and reactor ratios $M_Q(x)$ as function of travel distance for both the homogeneous and heterogeneous cases. It can be observed that for the homogeneous steady-state plume (Fig. 2a) the dilution index increases monotonically at a

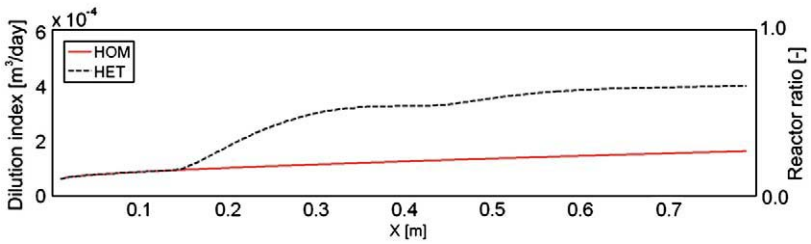

Fig. 5. Comparison between the computed dilution index and reactor ratio for the homogeneous and heterogeneous setups.

moderate rate along the flow direction. For the heterogeneous setup (Fig. 2b) the same behavior observed in the homogeneous system characterizes the portion of porous medium close to the inlet (same grain size). When the plume is focused into the first high-permeability inclusion, it is significantly diluted as can be noticed by the rapid increase of the dilution index. A plateau is reached between the two inclusions and further dilution takes place in the second zone. In this zone the increase of the dilution index is not as pronounced as in the first inclusion since the plume entering into the second high-permeability zone is already partially mixed with surrounding water. Using the flux-related dilution index as a measure of mixing, the degree of mixing of the fluorescein plume at the end of the heterogeneous tank is 2.46 times higher than in the homogeneous setup.

In the given application, the original volume-related dilution index would have been no useful measure of mixing, because we consider a steady-state concentration distribution of a continuously injected compound. This would yield a single value of an effective volume occupied by the plume. How the degree of mixing increases with travel distance could not be quantified. Likewise, modifying the dilution index to a cross-sectional quantity without weighting by the water flux would have failed to capture the physical process of transverse mixing taking place in the heterogeneous system. In fact, the shrinking of the plume in the high-permeability inclusion would have led to a decrease in the cross-section related dilution index at distances downstream of the high-permeability inclusions. This would have not been correct since, as mentioned above, dilution is an irreversible process and, under steady-state conditions, its measure should increase with the travel distance. Using the flux-related dilution index this physical constraint is fulfilled. In the high-permeability inclusions, the streamtubes get narrower leading to a smaller cross-sectional area, but transverse exchange makes the particles in the plume sample a larger fraction of the water flux.

We applied the concept of flux-related dilution index also to the transient experiments in the homogeneous setup. The results corresponding to two different oscillation frequencies are shown in Fig. 6. According to the computed values of dilution index, local mixing enhancements occur where the plume bends following the oscillating flow field. The mixing enhancements increase with time, as the signal propagates in the flow-through system. These findings, similar to the outcome of the second central spatial moment analysis, demonstrate that transient flow fields cause local mixing enhancements in comparison with the analogous steady-state setup.

3.2. Reactive experiments

3.2.1. Steady-state plumes

We performed reactive experiments by establishing an alkaline (0.004 mol/L NaOH) plume in an acidic ambient solution (0.01 mol/L HCl). Both solutions were amended with a concentration of 5.92×10^{-6} mol/L of bromophenol blue as pH indicator. The selected aqueous acid/base reaction is very fast, therefore it represents a suitable example for the investigation of mixing-controlled reactions in saturated porous media. The reaction between the acidic and alkaline solution reduces to water hydrolysis since the sodium and chloride ions behave as conservative compounds (spectator ions). The pH indicator added to both solutions behaves as a weak acid and undergoes the reaction $HIn \leftrightarrow H^+ + In^-$.

Titration experiments of 0.01 mol/L HCl solutions were performed with increasing volumes of alkaline solutions with different NaOH concentrations. The results plotted in Fig. 7 show the titration curves obtained with 0.004, 0.022 and 0.030 mol/L NaOH. The symbols (×) represent the measured pH whereas the continuous line shows the values computed with the speciation code PHREEQC-2 (Parkhurst and Appelo, 1999). X_{Alk} is the volumetric fraction of the alkaline solution in the mixture of the acidic and alkaline solutions, also denoted mixing ratio. A pH of 4.6, where the color change of the pH indicator is complete, was reached for mixing ratios of $X_{Alk}=0.712$, $X_{Alk}=0.309$, and $X_{Alk}=0.249$ (dashed lines) for the experiments with 0.004, 0.022 and 0.030 mol/L NaOH, respectively.

A steady-state alkaline plume was first established in the homogeneous tank by injecting the 0.004 mol/L NaOH solution in two adjacent central inlet ports and the ambient acidic solution (0.01 mol/L HCl) in the surrounding ports. The same procedure was repeated in the heterogeneous system. As shown in Fig. 8a and b, the plume shapes and lengths are quite different in the two systems: while in the homogeneous porous medium the plume extends for more than half the tank length, in the heterogeneous setup it appears to vanish as soon as it reaches the first high-permeability inclusion. As shown above for the conservative tracer, the focusing of flow causes an enhancement of mixing which leads, in the reactive case, to an enhanced acid/base reaction. This results in a considerably shorter alkaline plume. A successive increase of

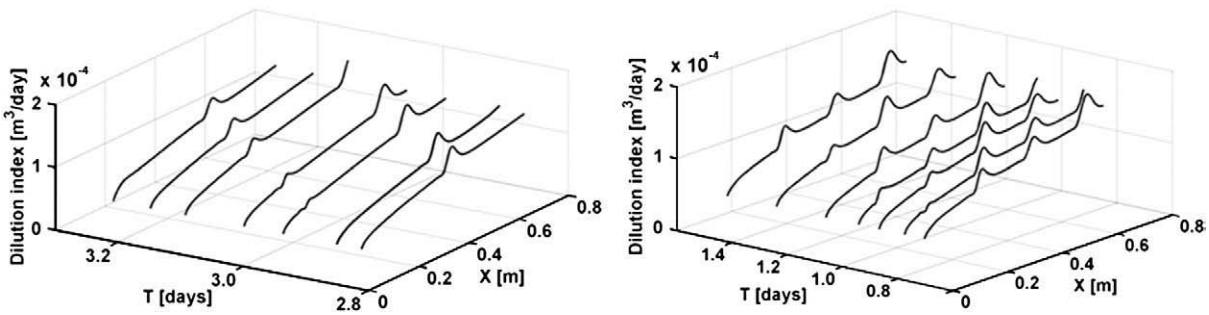

Fig. 6. Dilution index computed at different time in the transient setups, for the two distinct frequencies of flow oscillations.

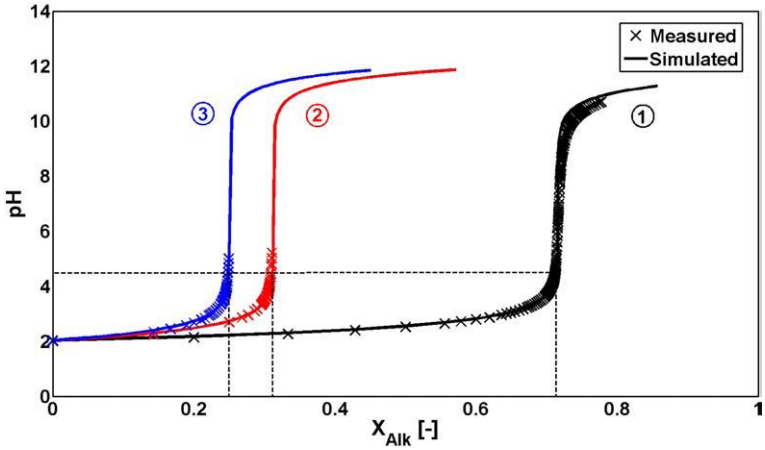

Fig. 7. Titration of 0.01 mol/L HCl solutions with alkaline solutions of 0.004, 0.022 and 0.030 mol/L NaOH (case 1, 2 and 3, respectively) using bromophenol blue (5.92×10^{-6} mol/L) as pH indicator.

the concentration of the inlet alkaline solution in the heterogeneous system resulted in longer plumes. For example, Fig. 8c shows a plume (0.022 mol/L NaOH) that travels through the first high-permeability inclusion and vanishes as it enters the second inclusion. Fig. 8d shows an alkaline plume (0.030 mol/L NaOH) traveling along the entire tank length.

We performed forward numerical modeling of the reactive experiments with the multi-component reactive transport code PHT3D (Prommer et al., 2003). The simulations were conducted using the flow and transport parameters summarized in Table 1, except for the transverse dispersivity that was estimated following Cirpka et al. (2006). According to this method, the

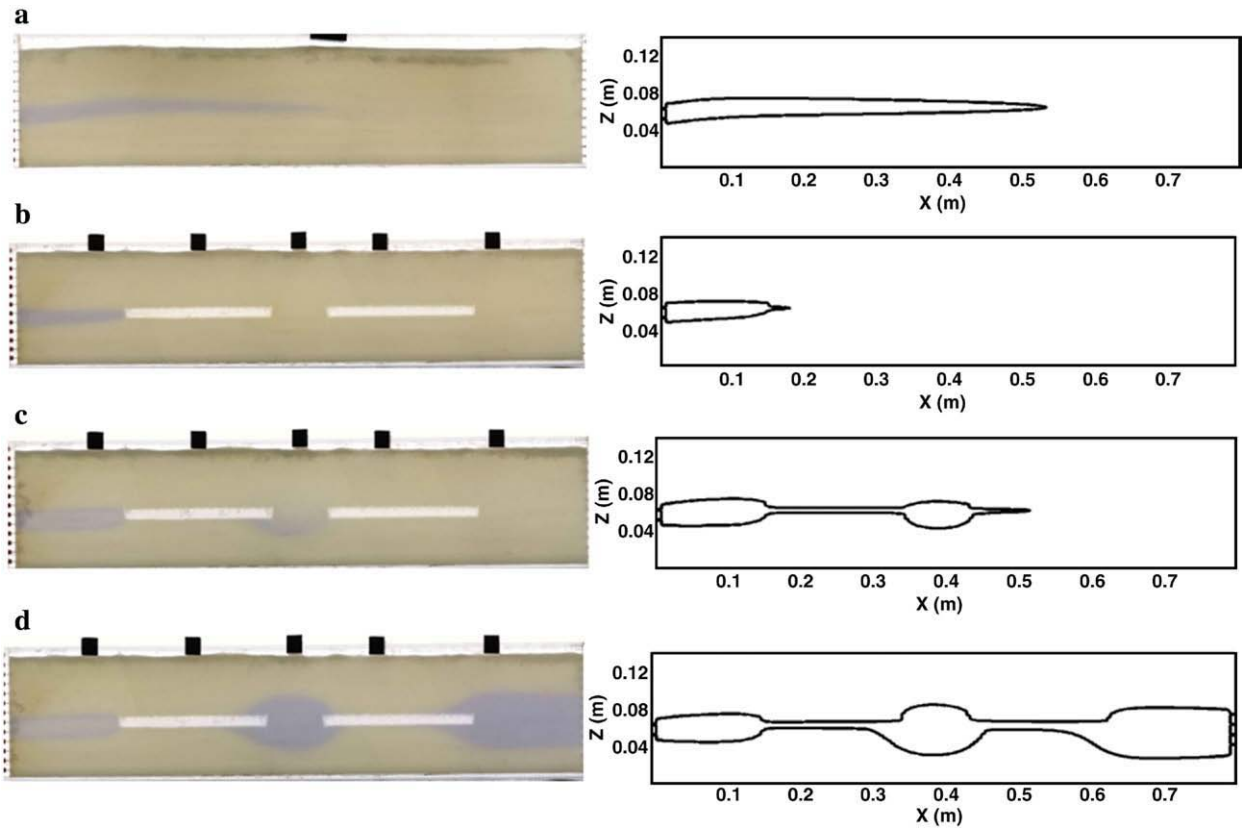

Fig. 8. Reactive plumes observed in the homogeneous and heterogeneous setup (left) and simulated with the numerical model PHT3D (contour lines corresponding to pH = 4.6). Plumes (a) and (b) were established injecting a concentration of 0.004 mol/L of NaOH, whereas (c) and (d) with 0.022 and 0.030 mol/L NaOH, respectively.

measured steady-state plume length in the homogeneous setup can be used to estimate the transverse dispersion coefficient by:

$$D_T = \frac{vY^2}{16L(inverf(X))^2} \tag{16}$$

where v is the seepage velocity, L is the plume length, Y is the injection height and X is the volume fraction corresponding to a specific pH value for which the plume can be visualized (pH = 4.6 in our application). Assuming a standard linear model for transverse dispersion, as explained above, transverse dispersivity can be estimated according to Eq. (3). The value of α_T (2.3×10^{-5} m) determined for the reactive experiments was of the same order of magnitude but higher than the one obtained in the conservative tracer experiments conducted in the same porous medium (1.4×10^{-5} m). The outcome of the reactive simulations are reported in Fig. 8 (right) and are consistent with the observed results.

3.2.2 . Analysis of reaction enhancement

Numerical simulations were performed in order to quantify the reaction enhancement in the heterogeneous tank in comparison to the homogeneous porous medium. The reactive system for this analysis was simplified to the instantaneous reaction between two dissolved reaction partners A and B giving the dissolved reaction product C:

$$A + B \rightarrow C. \tag{17}$$

The concentration of the reactants and product were calculated using the analytical solution of Cirpka and Valocchi (2007) after the numerical computation, in both the homogeneous and heterogeneous setups, of the transport of a conservative species (i.e. the volumetric mixing ratio of the reactants' solutions).

Fig. 9 shows the calculated mass fluxes of the product C of the instantaneous mixing-controlled reaction (Eq. (17)) taking place in the homogeneous and heterogeneous setups, where B is introduced in the central inlet port at a concentration ten times higher than that of reactant A (0.01 mol/L) in the ambient water. With these concentrations the plume of compound B reaches the outlet of the tank in both the homogeneous and heterogeneous porous media. The production of C can only take place at the fringe of the plume where reactants A and B mix by hydrodynamic transverse dispersion. The increasing trend of the product mass flux is similar to the one observed for the dilution index for a conservative compound. The mixing enhancement in the first high-permeability inclusion causes a significant reaction enhancement as can be noticed by the steep increase of the product mass flux. The influence of the second coarse grain inclusion is clearly less important and the rate of increase of the product mass flux is comparable, and almost parallel, to that observed in the homogeneous system. This result reflects that, after the first inclusion, the plume of B is spread out and its successive focusing into the second high-permeability zone is less effective. The flowlines carrying compound B are brought closely together inside the second inclusion, whereas the contact with the flowlines of the ambient groundwater, carrying the other reaction partner A, takes place mostly outside the coarse material. Therefore, the reactive zone turned out to be located outside the second inclusion and does not provide a significant reaction enhancement compared to the homogeneous setup. Other reactive transport simulations were performed assuming a setup similar to the experimental heterogeneous tank but with different locations of the second inclusion. Fig. 9a shows a scenario where the second high-permeability inclusion was shifted 3.2 cm towards the bottom of the tank. In Fig. 9b the second inclusion was divided into two thinner zones (each one with a thickness of half the experimental inclusion) located

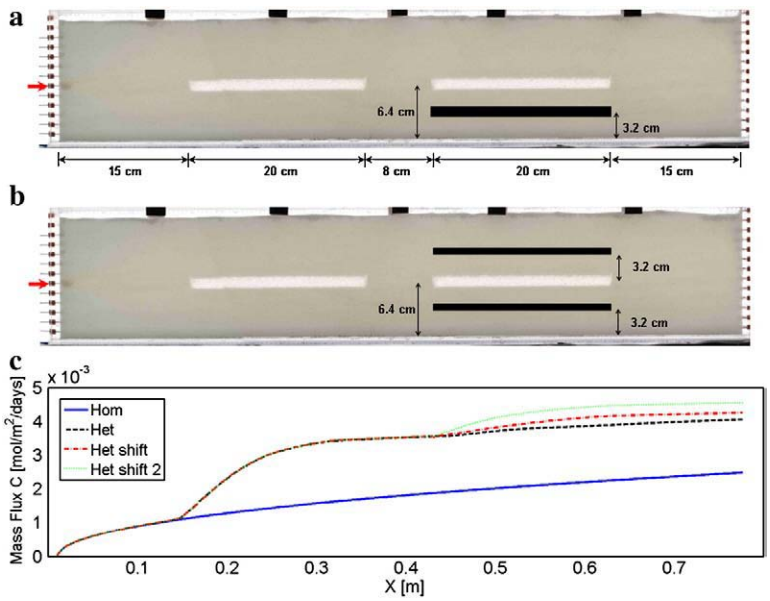

Fig. 9. Heterogeneous tank and virtual shift of the second high-permeability inclusion (a and b). Calculated product mass fluxes along the homogeneous and heterogeneous porous media (c).

Table 2
Product mass fluxes and reaction enhancement factors.

Setup	Inclusions	Flux C [mol/m^2/d]	RF
HOM — experimental	–	2.5×10^{-3}	1.00
HET — experimental	2 in line	4.1×10^{-3}	1.64
HET — simulated	2 shifted	4.3×10^{-3}	1.72
HET — simulated	3 shifted	4.5×10^{-3}	1.80

symmetrically with respect to the longitudinal axis of the inclusions in the experimental flow-through system. As can be noticed in Fig. 9c, the different location of the high-permeability inclusions resulted in different mass fluxes of the product. For the configuration shown in Fig. 9a, a higher product mass flux was calculated. This reaction enhancement is caused by the focusing of the lower fringe of the plume in the second high-permeability inclusion. In this case, both flowlines from the plume and from the surrounding ambient water converge into the second high-permeability zone resulting in a more effective mixing of A and B in the second half of the tank. The product mass flux additionally increases when the second inclusion is divided into two thinner zones (Fig. 9b), thus resulting in a more effective focusing of both the lower and the upper plume fringes.

The reaction enhancement factor (RF) was the parameter selected for the quantification of the reaction enhancement in the heterogeneous setups. RF is defined as the ratio between the mass flux of a reaction product in a heterogeneous porous medium and the product mass flux in the equivalent homogeneous system (Werth et al., 2006). In our application, we calculated the reaction enhancement factor, based on the outcome of the numerical flow and transport simulations, as:

$$RF = \frac{\int_0^{W_y}\int_0^{L_z} q_{\mathrm{HET}}(L,y,z)\cdot c_{\mathrm{C,HET}}(L,y,z)dydz}{\int_0^{W_y}\int_0^{L_z} q_{\mathrm{HOM}}(L,y,z)\cdot c_{\mathrm{C,HOM}}(L,y,z)dydz} \tag{18}$$

where $q_{\mathrm{HET}}(L,y,z)$ and $q_{\mathrm{HOM}}(L,y,z)$ are the water fluxes computed by the flow model at the outlet cells of the heterogeneous and homogeneous domains, respectively; $c_{\mathrm{C,HET}}(L,y,z)$ and $c_{\mathrm{C,HOM}}(L,y,z)$ are the concentrations of the product at the outlet.

The reaction enhancement factor in the heterogeneous experimental setup was calculated to be 1.64. This means that the focusing of flow in the high-permeability inclusions and its consequent increase of mixing result in considerable higher mass fluxes of a product of an instantaneous, mixing-controlled reaction in the heterogeneous tank compared to the analogous homogeneous porous medium. An increase of RF from 1.64 to 1.72 was obtained when the second coarse grain size zone was shifted (Fig. 9a), therefore resulting in a more efficient focusing of the lower reactive fringe of the plume in the second high-permeability inclusion (Table 2). A higher increase of RF (8.9%), in comparison with the experimental heterogeneous setup, was calculated for the configurations shown in Fig. 9b.

Reactive experiments were conducted also under transient flow conditions. In analogy to the conservative transport experiments using fluorescein, an alkaline plume was established in the homogeneous porous medium under oscillating flow conditions. The oscillating flow field was induced by using different flow rates at the inlet and outlet ports as described above. Pictures of the alkaline plume were taken at different times. The oscillating plumes generally showed a plume length comparable to the one of an analogous experiment in the same homogeneous setup but under steady-state conditions. Snapshots of the transient plumes after 1.6 days (a) and 2.1 days (b) from the beginning of the experiment are shown in Fig. 10, together with the simulated contours at pH = 4.6.

A quantitative comparison of reaction enhancement between the transient and steady-state experiments was done on the basis of the outcome of numerical simulations. In analogy with the analysis carried out in the heterogeneous setup, the instantaneous reaction (Eq. (17)), with injection of the same alkaline and acidic solutions, was considered. Thus, an alkaline transient plume stretching over the whole domain was obtained. The mass fluxes of the product C, generated by mixing and reactions at the plume fringe, were calculated along the length of the flow-through system. Fig. 11 shows spatial profiles of the product mass flux in the homogeneous tank under steady-state and transient conditions, respectively. Similarly to what was observed by calculating the second central spatial moment of a conservative tracer (Fig. 3c and d), the mass flux of the reaction product shows local increase compared to the steady-state setup. In fact, the profile taken under transient conditions shows mixing enhancement corresponding to the points were the reactive plume bends as a consequence of the flow field induced by the pumps' switch.

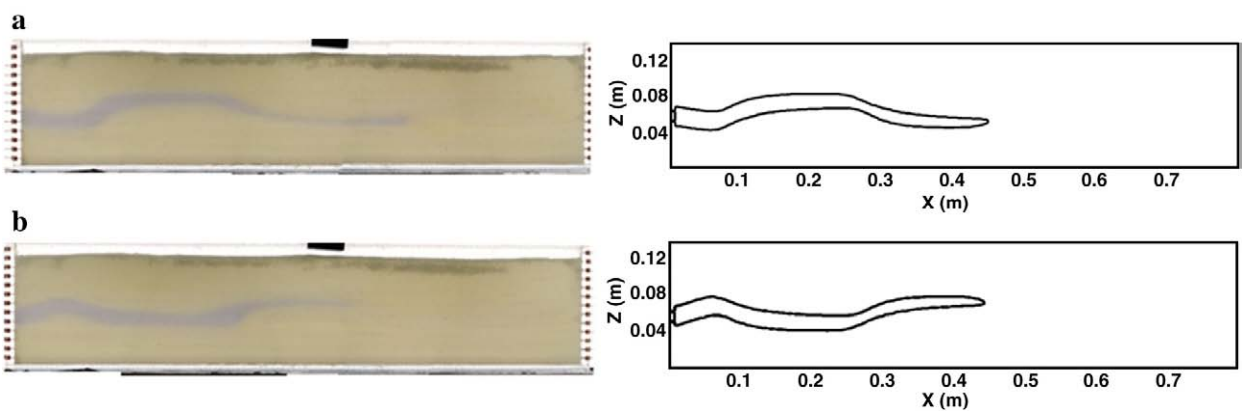

Fig. 10. Observed and simulated (contour at pH = 4.6) transient reactive plumes.

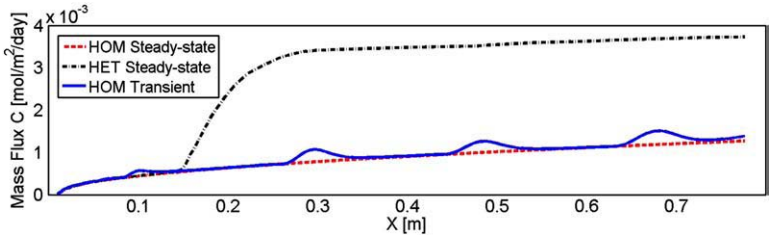

Fig. 11. Product mass fluxes in the homogeneous tank under steady-state (dash line) and transient conditions (solid lines). The dash–dot line represents the product mass flux in the heterogeneous system at steady-state.

However, these local mixing and reaction enhancements are less relevant compared to the ones computed for the heterogeneous system (dash–dot line).

4. Conclusions

In this paper we experimentally investigated the effects of physical heterogeneity and transient flow conditions on mixing and mixing-controlled reactions in saturated porous media. The laboratory experiments were performed using both conservative and reactive tracers and the interpretation of the measured data and observed plume behavior was aided by forward numerical modeling. Conservative and reactive non-sorbing compounds were selected in order to limit the investigation to mixing caused by transverse hydrodynamic dispersion.

Mixing and reaction enhancements were caused by both transient flow fields and flow focusing into high-permeability inclusions compared to analogous homogeneous setups under steady-state flow conditions. However, for both conservative and reactive tracers, the contribution of porous medium heterogeneity, in the form of geometrically well-defined high-permeability inclusions, was distinctly more significant. The results of this study demonstrate the importance of flow focusing, suggested in the theoretical work of Werth et al. (2006), for the development of dissolved plumes in groundwater. For plumes whose behavior is controlled by diffusive/dispersive mixing-controlled reactions at the plume fringe, the amount of flow focusing, caused by the permeability contrast between the different materials, is of utmost importance in determining the plume development and length. Moreover, the extent to which the reactive fringe is focused considerably contributes to the degree of mixing enhancement. This depends on the geometry of the porous medium and, in particular, on the location of high-permeability inclusions.

For conservative and reactive non-sorbing compounds, oscillating flow fields only cause local mixing enhancements where the groundwater flow changes its principal direction. For the investigated settings, these enhancements were significantly lower than those obtained by plume focusing and mixing in the high-permeability inclusions.

Forward conservative and reactive transport modeling showed a good agreement with the measured concentrations and the observed behavior of the plumes. Based on the outcome of the numerical simulations, a quantitative analysis of the different experiments was possible. The transverse spatial moments of the plume were calculated for the conservative experiments, whereas for the reactive experiments the quantitative analysis was based on the computed fluxes of the product of an instantaneous mixing-controlled reaction. In particular, the definition of a flux-related dilution index proved to be of significant importance for quantifying mixing in both homogeneous and heterogeneous porous media. In the latter systems the second central spatial moments fail to describe plume mixing and a clear distinction between plume spreading and dilution is necessary.

Although the absolute values of the computed parameters are likely to be specific for the simplified experimental settings used in our study; important outcomes have a broader significance. For instance, the analysis based on the dilution index for the conservative tracer and on product mass fluxes for the reactive experiments, clearly shows the major contribution of the first high-permeability inclusion on mixing and reaction enhancement in the heterogeneous setup. In this zone, the rate of dilution increases considerably, as an “undiluted plume”, characterized by sharp gradients at the fringes, is effectively mixed with the surrounding clean water. These findings suggest the importance of focusing of the reactive fringe close to contamination sources for groundwater plumes. For organic contaminant plumes undergoing fringe biodegradation processes, the microbial activity is expected to be enhanced in high-permeability zones located close to contaminant sources and hot-spots, provided that the high contaminant concentrations do not inhibit the activity of the microorganisms and that ambient groundwater, rich in dissolved electron acceptors, is also focused into the high-permeability inclusions. This behavior was observed in experiments on biodegradation of organic contaminants in flow-through laboratory experiments (Cirpka et al., 1999; Bauer et al., 2009a and Bauer et al., 2009b).

Oscillating flow fields, cause only limited and local mixing enhancements for conservative and reactive non-sorbing tracers as considered in our experimental investigation. Some previous modeling studies (e.g. Prommer et al., 2002; Cirpka, 2005) suggest a more significant one-time chromatographic mixing event for sorbing organic compounds. However, the mixing enhancement can entail a decrease of the overall contaminant degradation if the microbial populations mediating the transformations are not able to react sufficiently fast to the changes of environmental conditions caused by transient flow.

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